

# CS & IT ENGINEERING



## Operating System

### Deadlock

Lecture – 02

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# Recap of Previous Lecture



**Topic**

**Reader-Writer Problem**

**Topic**

**Dining-Philosopher Problem**



# Topics to be Covered



**Topic**

**Deadlock Prevention**

**Topic**

**Deadlock Avoidance**



## Topic : Recovery From Deadlock

1. Make Sure that deadlock never occur
  - Prevent the system from deadlock or avoid deadlock
2. Allow deadlock, detect and recover
3. Pretend that there is no any deadlock  $\Rightarrow$  all latest OS



## Topic : Deadlock Prevention

↓  
prevent atleast one of the 4 necessary Conditions





## Topic : Preventing Mutual Exclusion

1. Either make all processes independent
2. Avail seperate set of resources for each process.

not practically possible



## Topic : Preventing Hold and Wait

→ Each process should either hold all required resources  
or should wait for all required resources

Problem:-

1. A process may suffer from starvation
2. Not appropriate utilization of resources



## Topic : Preventing No-Preemption

↓  
allow preemption of resources

1. Process from which resource preempted, can go to unstable state
2. selection of a victim process
3. victim process may suffer from starvation

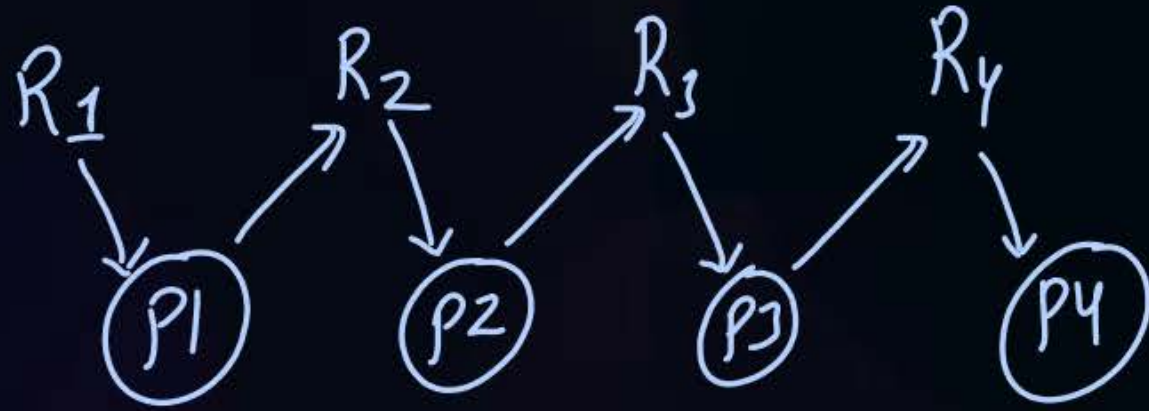




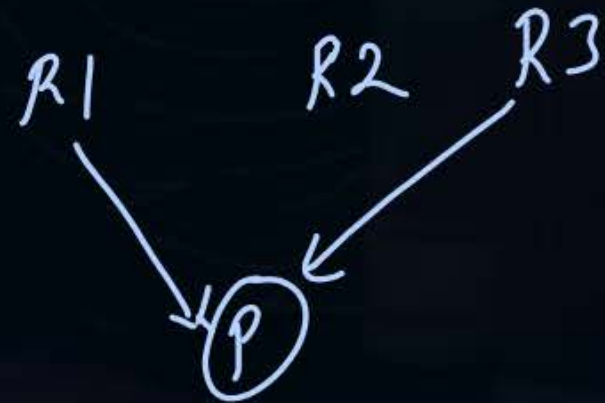
## Topic : Preventing Circular Wait

Sol<sup>n</sup>:-

Give number to each resource  $R_1, R_2, \dots$  when a process holds a resource  $R_i$  and wants to request resource  $R_j$ , then it is allowed only when  $j > i$ .



If a process  $p$  holds  $R_1, R_3$  and needs  $R_2 \Rightarrow$  Release  $R_3$  then try to acquire resource  $R_2$  then  $R_3$ .





## Topic : Deadlock Avoidance

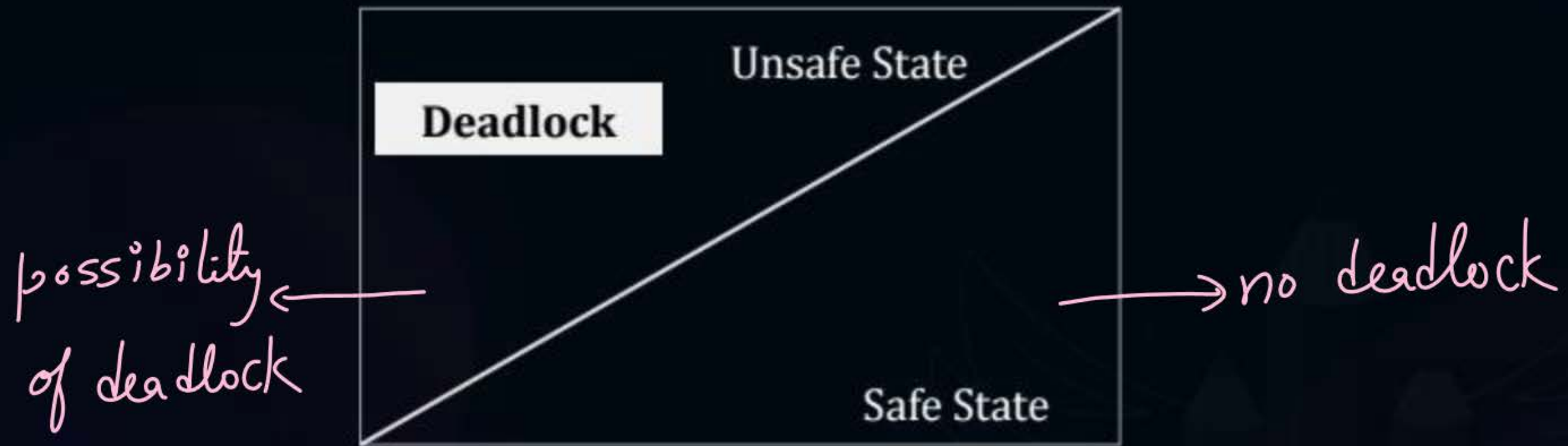
In deadlock avoidance, the OS tries to keep system in safe state

↓  
OS allocates resources to processes  
only when OS makes sure that there is  
no any deadlock.



## Topic : Deadlock Avoidance

In deadlock avoidance, the OS tries to keep system in safe state







## Topic : Deadlock Avoidance

In deadlock avoidance, the request for any resource will be granted if the resulting state of the system doesn't cause deadlock in the system.



## Topic : Banker's Algorithm

↳ bank

The banker's algorithm is a resource allocation and deadlock avoidance algorithm that tests for safety

Banker's algo  
→ safety algorithm ⇒ system safe or not

→ Resource - Request Algorithm

whenever a process arrives  
each must declare to  
OS about its max requirement  
of resources.





## Topic : Banker's Algorithm

already allocated

safety algo  
with OS

$Need = Max - allocation$

| Process | Allocation | Max | Available                                                       |
|---------|------------|-----|-----------------------------------------------------------------|
| P1      | 1          | 3   | 1                                                               |
| P2      | 5          | 8   | After $1+3=4$<br>P3                                             |
| P3      | 3          | 4   | After $\Rightarrow 4+1=5$<br>P1                                 |
| P4      | 2          | 7   | After P2 $\Rightarrow 5+5=10$<br>After P4 $\Rightarrow 10+2=12$ |

2

3

1

5

find a process  $p_i$ , with  $need_i \leq available \Rightarrow P3 \Rightarrow P3$  completes

$Available = Available + Allocation_i$

$\Downarrow$

if all processes finished  $\Rightarrow$  system is safe



safe sequence  $\Rightarrow \langle P_3, P_1, P_2, P_4 \rangle$



## Topic : Banker's Algorithm

A, B, C  $\Rightarrow$  3 types of resources

| Process        | Allocation | Max   | Available              | Need               |
|----------------|------------|-------|------------------------|--------------------|
|                | A B C      | A B C | A B C                  | A B C              |
| P <sub>0</sub> | 0 1 0      | 7 5 3 | 3 3 2                  | 7 4 3              |
| P <sub>1</sub> | 2 0 0      | 3 2 2 | $\checkmark$ P1 5 3 2  | 1 2 2 $\checkmark$ |
| P <sub>2</sub> | 3 0 2      | 9 0 2 | $\checkmark$ P3 7 4 3  | 6 0 0              |
| P <sub>3</sub> | 2 1 1      | 2 2 2 | $\checkmark$ P0 7 5 3  | 0 1 1 $\checkmark$ |
| P <sub>4</sub> | 0 0 2      | 4 3 3 | $\checkmark$ P2 10 5 5 | 4 3 1              |
|                |            |       | $\checkmark$ P4 10 5 7 |                    |

system safe

safe sequence  $\Rightarrow \langle P_1, P_3, P_0, P_2, P_4 \rangle$

$\neq$  seq.  $\Rightarrow$  safe seq. or not

1.  $\langle P_2, P_4, P_0, P_1, P_3 \rangle \Rightarrow$  Not safe

2.)  $\langle P_3, P_2, P_1, P_0, P_4 \rangle \Rightarrow$  Not safe

3)  $\langle P_1, P_4, P_0, P_2, P_3 \rangle \Rightarrow$  Not safe

4)  $\langle P_4, P_0, P_1, P_2, P_3 \rangle \Rightarrow$  Not safe



H.W. Safe or not ?

#Q.

| Process | Allocation<br>A B C D | Max<br>A B C D | Available<br>A B C D |
|---------|-----------------------|----------------|----------------------|
| P1      | 0 0 1 2               | 0 0 1 2        | 1 5 2 0              |
| P2      | 1 0 0 0               | 1 7 5 0        |                      |
| P3      | 1 3 5 4               | 2 3 5 6        |                      |
| P4      | 0 6 3 2               | 0 6 5 4        |                      |
| P5      | 0 0 1 4               | 0 6 5 6        |                      |



## 2 mins Summary

**Topic**

**Deadlock Prevention**

**Topic**

**Deadlock Avoidance**



**Happy Learning**

**THANK - YOU**